

Probability
Chapter 2 - Module 4

Moment Generating Functions

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Def. (Moment Generating Function) Let X be a real valued random variable. The moment generating function for X is

$$\phi(t) = E \left[e^{tX} \right] = \begin{cases} \sum_x e^{tx} p(x), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx, & \text{if } X \text{ is continuous,} \end{cases}$$

whenever these sums and integrals exists.

In order for the formula to be useful, there must be some number h so that $\phi(t)$ is well define on $[-h, h]$.

We can see how $\phi(t)$ generates moments in the follow way.

If X is discrete and $\phi(t)$ is differentiable, then formally

$$\phi'(t) = \frac{d}{dt} \sum_x e^{tx} p(x) = \sum_x x e^{tx} p(x)$$

and so

$$\phi'(0) = \sum_x x p(x),$$

which is the first moment of X .

Similarly

$$\phi''(t) = \frac{d^2}{dt^2} \sum_x e^{tx} p(x) = \sum_x x^2 e^{tx} p(x)$$

and so

$$\phi''(0) = \sum_x x^2 p(x),$$

which is the second moment of X .

In general $\phi^{(n)}(0) = \sum_x x^n p(x)$, which is the n -th moment of X .

If X is a continuous random variable and $\phi(t)$ is differentiable, then formally

$$\phi'(t) = \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^{\infty} x e^{tx} f(x) dx$$

and so

$$\phi'(0) = \int_{-\infty}^{\infty} x f(x) dx,$$

which is the first moment of X .

Similarly

$$\phi''(t) = \frac{d^2}{dt^2} \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^{\infty} x^2 e^{tx} f(x) dx$$

and so

$$\phi''(0) = \int_{-\infty}^{\infty} x^2 f(x) dx,$$

which is the second moment of X .

In general $\phi^{(n)}(0) = \int_{-\infty}^{\infty} x^n f(x) dx$, which is the n -th moment of X .

Example 1. (Example 2.40) The Binomial distribution with parameters n and p , has probabilities

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

for $k = 0, 1, \dots, n$.

Then

$$\begin{aligned}\phi(t) &= \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} (pe^t)^k (1-p)^{n-k} \\ &= (pe^t + 1 - p)^n.\end{aligned}$$

Hence

$$\phi'(t) = n(pe^t + 1 - p)^{n-1} pe^t$$

and so

$$E[X] = \phi'(0) = np.$$

Furthermore, since

$$\phi'(t) = n(pe^t + 1 - p)^{n-1}pe^t,$$

then

$$\phi''(t) = n(n-1)(pe^t + 1 - p)^{n-2}(pe^t)^2 + n(pe^t + 1 - p)^{n-1}pe^t$$

and so

$$E[X^2] = \phi''(0) = n(n-1)p^2 + np.$$

Therefore

$$\text{Var}(X) = E[X^2] - (E[X])^2 = n(n-1)p^2 + np - n^2p^2 = np(1-p).$$

Example 2. (Ex. 2.43) Suppose that Z is a normally distributed random variable with mean 0 and variance 1. Then

$$\begin{aligned}\phi(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} e^{xt} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2-2xt)/2} dx \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx = e^{t^2/2}\end{aligned}$$

Now let $X = \sigma Z + \mu$ so that $X \sim N(\mu, \sigma)$. Then

$$\phi(t) = E[e^{tX}] = E[e^{t(\sigma Z + \mu)}] = e^{t\mu} E[e^{(t\sigma)Z}] = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

A simple calculations shows that $\phi'(0) = \mu$, $\phi''(0) = \mu^2 + \sigma^2$. So that $\text{Var}(X) = E[X^2] - (E[X])^2 = \sigma^2$. (Book has a “(” in the wrong place on page 61.)

Example 3. Let X be uniformly distributed on the interval $[0, 1]$. Then

$$\phi(t) = E[e^{Xt}] = \int_{-\infty}^{\infty} e^{xt} f(x) dx = \int_0^1 e^{xt} dx = \frac{1}{t} e^{xt} \Big|_{x=0}^{x=1} = \frac{1}{t} (e^t - 1).$$

But clearly we cannot let $t = 0$. So write

$$\begin{aligned} \phi(t) &= \frac{1}{t} (e^t - 1) \\ &= \frac{1}{t} [(1 + t + t^2/2 + t^3/3! + t^4/4! + \cdots) - 1] \\ &= 1 + t/2 + t^2/3! + t^3/4! + \cdots + t^n/(n+1)! + \cdots \end{aligned}$$

Then it is not too difficult to see that $\phi'(0) = 1/2$, $\phi''(0) = 1/3$ and that in general, $\phi^{(n)}(0) = 1/(n+1)$. And it is also easy to check this by direct integration.

Note finally that $\text{Var}(X) = E[X^2] - (E[X])^2 = 1/3 - 1/4 = 1/12$.

There are many important facts that will be useful to us later.

Theorem. If the moment generating function is defined for $-h \leq t \leq h$ for some positive number h , then it uniquely determines the distribution.

So if $E[e^{Xt}] = E[e^{Yt}]$ for two random variable X and Y , then they have the same distribution.

Theorem. If X and Y are independent random variables then

$$E \left[e^{(X+Y)t} \right] = E \left[e^{Xt} \right] E \left[e^{Yt} \right]$$

Proof. The proof is easy. We only need to note that

$$E \left[e^{(X+Y)t} \right] = E \left[e^{Xt} e^{Yt} \right] = E \left[e^{Xt} \right] E \left[e^{Yt} \right]$$

since if X and Y are independent, then so is $g(X)$ and $g(Y)$ for any function g .

We will use both of these facts to prove the following theorem.

Theorem. Let $X \sim N(\mu_1, \sigma_1)$ and $Y \sim N(\mu_2, \sigma_2)$ and suppose that X and Y are independent. Then $X + Y$ is normally distributed with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$.

Proof.

$$\begin{aligned}\phi_{X+Y}(t) &= \phi_X(t)\phi_Y(t) \\ &= \exp\left(\frac{\sigma_1^2 t^2}{2} + \mu_1 t\right) \exp\left(\frac{\sigma_2^2 t^2}{2} + \mu_2 t\right) \\ &= \exp\left(\frac{\sigma_1^2 t^2}{2} + \frac{\sigma_2^2 t^2}{2} + \mu_1 t + \mu_2 t\right) \\ &= \exp\left(\frac{(\sigma_1^2 + \sigma_2^2)t^2}{2} + (\mu_1 + \mu_2)t\right),\end{aligned}$$

which is the moment generating function for a normal distribution with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$!

Def. We say that $X \in \mathcal{R}^n$ is a multivariable normal random variable iff for some $\mu \in \mathcal{R}^n$ and some positive definite symmetrix matrix Σ , that

$$f(x) = \frac{1}{(\sqrt{2\pi})^n} |\Sigma|^{-1/2} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}.$$

Note both x and μ are column vectors of length n .

We will develop this from first principles, but note that when $n = 1$, then we should write $\Sigma = \sigma^2$ to get the one dimensional normal density function

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}.$$

So let $Z_1, Z_2, \dots, Z_n$ be independent identically distributed normal random variables with mean 0 and variance 1.

Then the joint pdf of $Z_1, Z_2, \dots, Z_n$ is

$$\begin{aligned} f_Z(z_1, z_2, \dots, z_n) &= f(z_1)f(z_2) \cdots f(z_n) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-z_i^2/2}. \\ &= \left(\frac{1}{\sqrt{2\pi}} \right)^n e^{-\sum_{i=1}^n z_i^2/2}. \\ &= \left(\frac{1}{\sqrt{2\pi}} \right)^n e^{-\frac{1}{2} z^T z}. \end{aligned}$$

So now consider the linear transformation

$$\begin{aligned}X_1 &= a_{11}Z_1 + a_{12}Z_2 + \cdots + a_{1n}Z_n + \mu_1 \\X_2 &= a_{21}Z_1 + a_{22}Z_2 + \cdots + a_{2n}Z_n + \mu_2 \\&\vdots = \vdots \\X_n &= a_{n1}Z_1 + a_{n2}Z_2 + \cdots + a_{nn}Z_n + \mu_n\end{aligned}$$

Then if we define

$$x = g(z) \equiv Az + \mu,$$

then the Jacobian is

$$J(z) = |A|$$

and the inverse of g is

$$z = h(x) \equiv A^{-1}(x - \mu).$$

These are all of the pieces that we need and the distribution for

$$X = g(Z) = AZ + \mu$$

is

$$\begin{aligned} f_X(x) &= f_Z(z)|J(z)|^{-1} \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^n e^{-\frac{1}{2}z^T z} |J(z)|^{-1} \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^n e^{-\frac{1}{2}(x-\mu)^T A^{-T} A^{-1}(x-\mu)} |A|^{-1} \end{aligned}$$

since $z = A^{-1}(x - \mu)$ and $z^T = (x - \mu)^T A^{-T}$.

To put this in a more traditional form define

$$\Sigma = AA^T.$$

Then $|\Sigma| = |AA^T| = |A| |A^T| = |A| |A| = |A|^2$ and so $|A| = |\Sigma|^{1/2}$.

And also

$$\Sigma^{-1} = \left(AA^T \right)^{-1} = A^{-T} A^{-1}.$$

With these substitutions we get

$$f_X(x) = \left(\frac{1}{\sqrt{2\pi}} \right)^n |\Sigma|^{-1/2} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}.$$

Question: So what is Σ ?

Consider

$$X_1 = a_{11}Z_1 + a_{12}Z_2 + \cdots + a_{1n}Z_n + \mu_1.$$

Then

$$\begin{aligned}\text{Var}(X_1) &= E[(X_1 - E[X_1])^2] \\ &= E[(X_1 - \mu_1)^2] \\ &= E[(a_{11}Z_1 + a_{12}Z_2 + \cdots + a_{1n}Z_n)^2] \\ &= \text{Var}[a_{11}Z_1 + a_{12}Z_2 + \cdots + a_{1n}Z_n] \\ &= a_{11}^2 \text{Var}[Z_1] + a_{12}^2 \text{Var}(Z_2) + \cdots + a_{1n}^2 \text{Var}(Z_n) \\ &= a_{11}^2 + a_{12}^2 + \cdots + a_{1n}^2 \\ &= \Sigma_{11}.\end{aligned}$$

Similarly

$$\Sigma_{ii} = \text{Var}(X_i) \equiv \sigma_i^2$$

and

$$\Sigma_{ij} = \Sigma_{ji} = \text{Cov}(X_i, X_j) = \sigma_i \sigma_j \rho(X_i, X_j).$$

Def. (n-dimensional Moment Generating Function) The moment generating function for $X = \{X_1, X_2, \dots, X_n\}$ is

$$\phi(t_1, t_2, \dots, t_n) = E \left[e^{\sum_{i=1}^n t_i X_i} \right] = E \left[e^{t^T X} \right].$$

If $X \sim N(\mu, \Sigma)$, then

$$\phi(t_1, t_2, \dots, t_n) = \exp \left(\sum_{i=1}^n t_i \mu_i + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n t_i t_j \text{Cov}(X_i, X_j) \right).$$

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed normal random variable with mean μ and variance σ^2 .

We consider the following two statistics:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

and

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

We easily note that

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu.$$

With a little more work we will show that

$$E[S^2] = \sigma^2.$$

We first note that

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

So

$$\begin{aligned} E[(n-1)S^2] &= \sum_{i=1}^n E[(x_i - \mu)^2] - nE[(\bar{X} - \mu)^2] \\ &= n\sigma^2 - n\text{Var}(\bar{X}) \\ &= (n-1)\sigma^2. \end{aligned}$$

In addition, we note that $\bar{X}$ is independent of $X_1 - \bar{X}, X_2 - \bar{X}, \dots, X_n - \bar{X}$, which implies that

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is independent of $\bar{X}$.

So what is the distribution of S^2 ?

Def. (Chi-Squared Distribution) Let $Z_1, Z_2, \dots, Z_n$ be independent standard normal random variables. Let

$$Y = Z_1^2 + Z_2^2 + \dots + Z_n^2.$$

The distribution for Y by definition is the Chi-Squared distribution with n degrees of freedom.

Let's begin to understand this by deriving the moment generating function for Y . So we need to determine

$$\begin{aligned} E[e^{tY}] &= E\left[e^{t\sum_{i=1}^n Z_i^2}\right] \\ &= \prod_{i=1}^n E\left[e^{tZ_i^2}\right], \end{aligned}$$

using the fact that $Z_1, Z_2, \dots, Z_n$ are all independent.

So let Z be a standard normal random variable.

Let $\sigma^2 = (1 - 2t)^{-1}$. Then

$$\begin{aligned} E \left[e^{tZ^2} \right] &= \int_{-\infty}^{\infty} e^{tx^2} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{tx^2 - x^2/2} dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x^2/2)(1-2t)} dx \\ &= \sigma \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} e^{-(x^2/2\sigma^2)} dx \\ &= \sigma = (1 - 2t)^{-1/2} \end{aligned}$$

Note that $\sigma^2 = (1 - 2t)^{-1}$ as long as $|t| < \frac{1}{2}$.

So

$$E \left[e^{tZ^2} \right] = (1 - 2t)^{-1/2}$$

and so

$$E \left[e^{t \sum_{i=1}^n Z_i^2} \right] = \prod_{i=1}^n E \left[e^{tZ_i^2} \right] = (1 - 2t)^{-n/2}.$$

This is the moment generating function for

$$\chi^2(n) \equiv \begin{cases} \frac{1}{2^{n/2}} \frac{1}{\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-\frac{x}{2}} & x \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

where

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$$

and where

$$\Gamma(k) = (k-1)!$$

when k is a positive integer.

So what about the distribution of

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 ?$$

Note (or recall) that

$$(n-1)S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

so that

$$(n-1) \frac{S^2}{\sigma^2} + n \left(\frac{\bar{X} - \mu}{\sigma} \right)^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2.$$

and so

$$(n-1) \frac{S^2}{\sigma^2} + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2.$$

The second term is $\chi^2(1)$ and the term on the right is $\chi^2(n)$. So using moment generating functions we can show that

$$(n-1) \frac{S^2}{\sigma^2}$$

is a χ^2 random variable with $n-1$ degrees of freedom.

To summarize:

Theorem. Let $X_1, X_2, \dots, X_n$ be iid, $N(\mu, \sigma)$ random variables.

Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

and

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

and where $\bar{X}$ and S^2 are independent.